Digitalizing a nurse handover tool

By Group 10: Ziwei, Sapnna, Miriam, Ohene, Grace

CONTENTS

01 Problem statement

02 Prototype

03 Design principles

04 Technical architecture

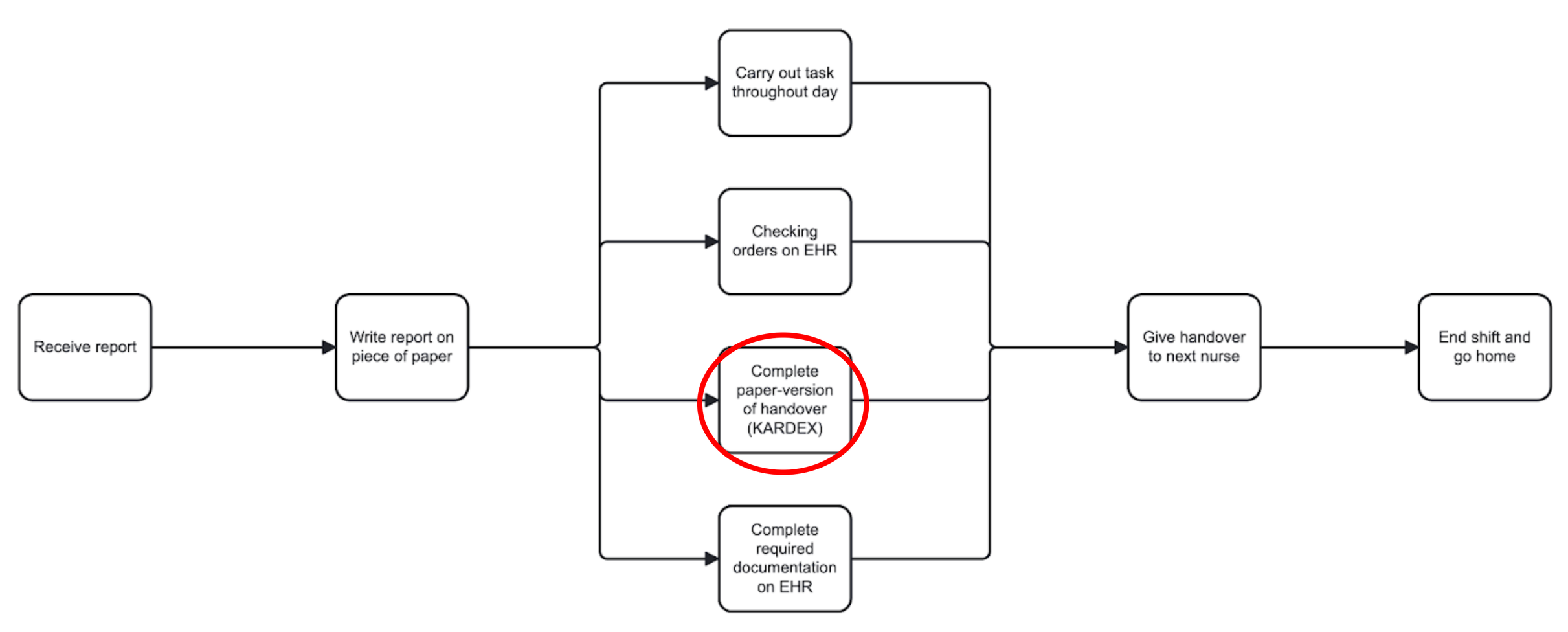
05 Business Case

INTRODUCTION

Why is shift handover an important process for nurses?

- Critical moments of communication
- Plans of care
- Prioritization
- Direct patient outcome

Current Workflow



Why this paper tool a problem?

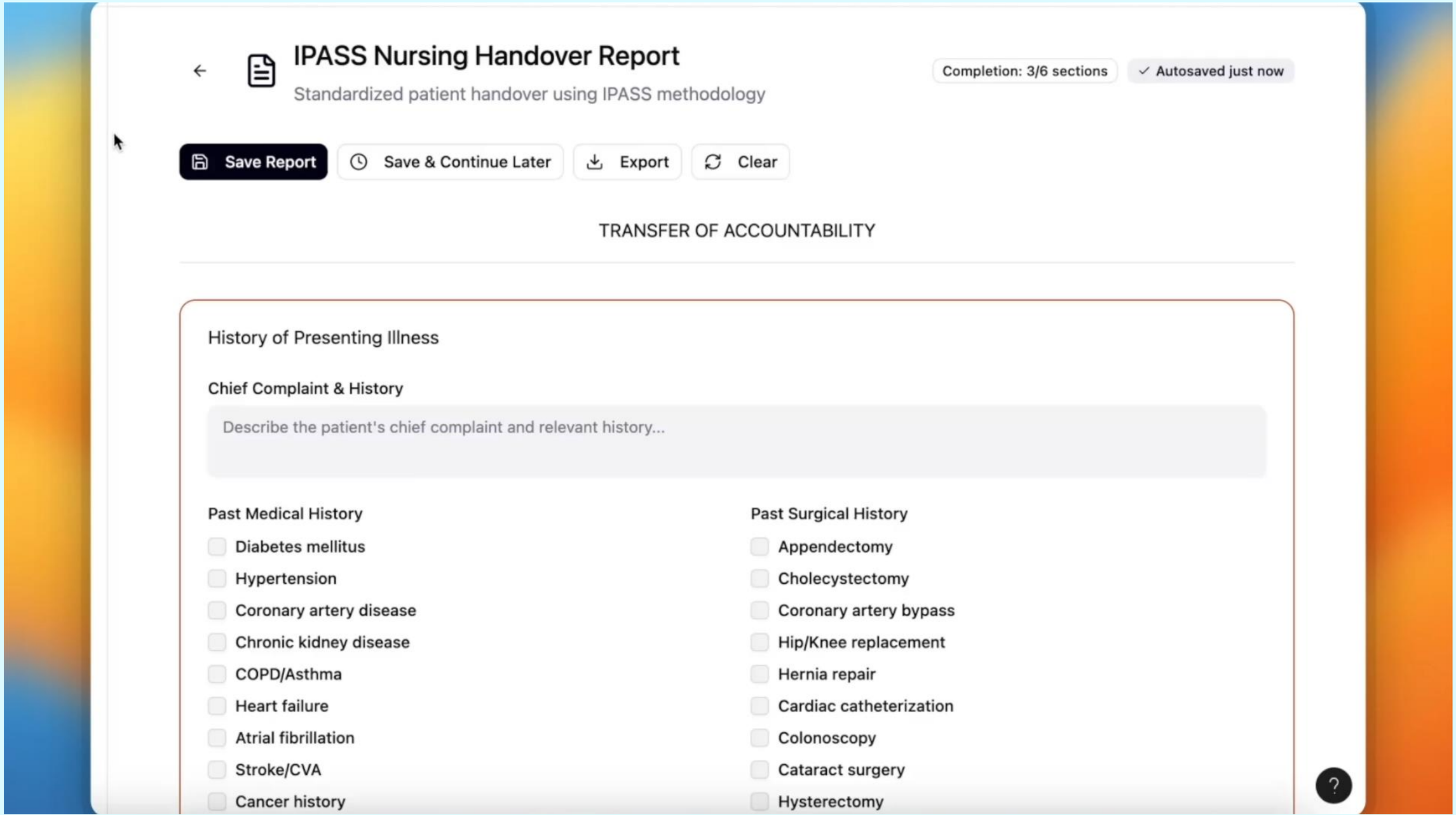
Redundancy

Inconsistency

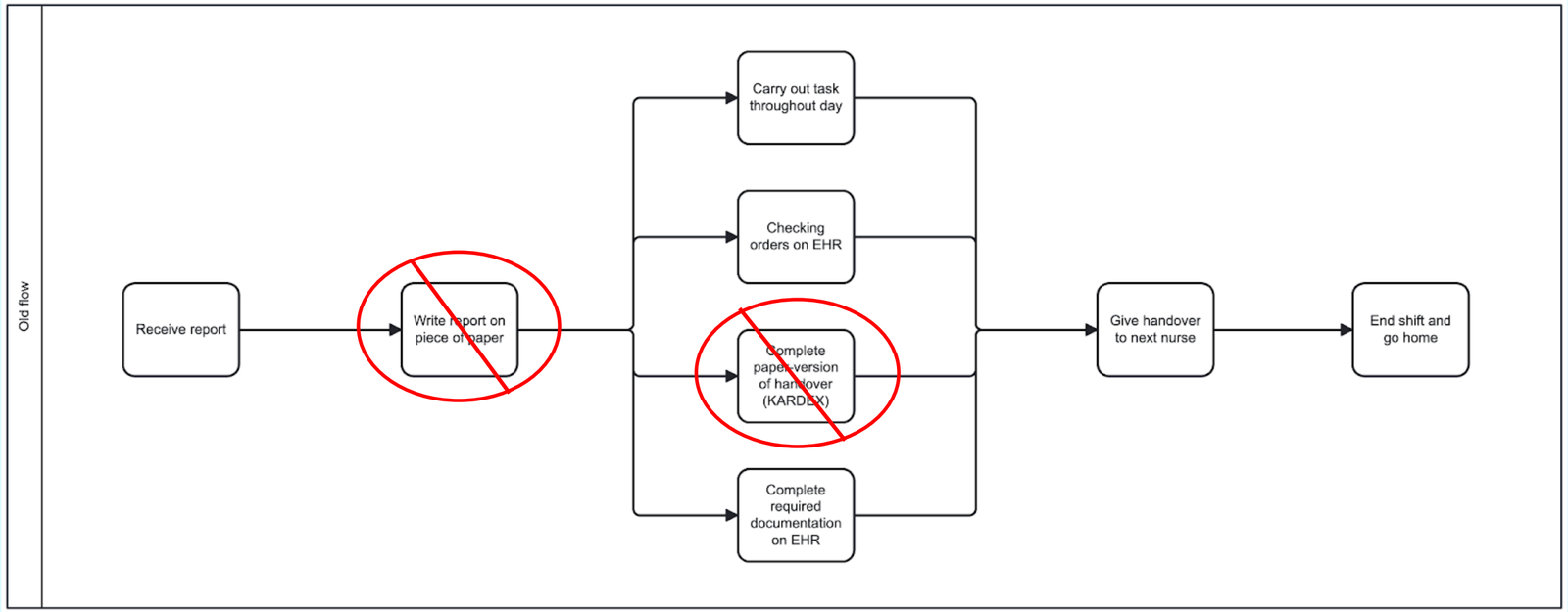
Cost

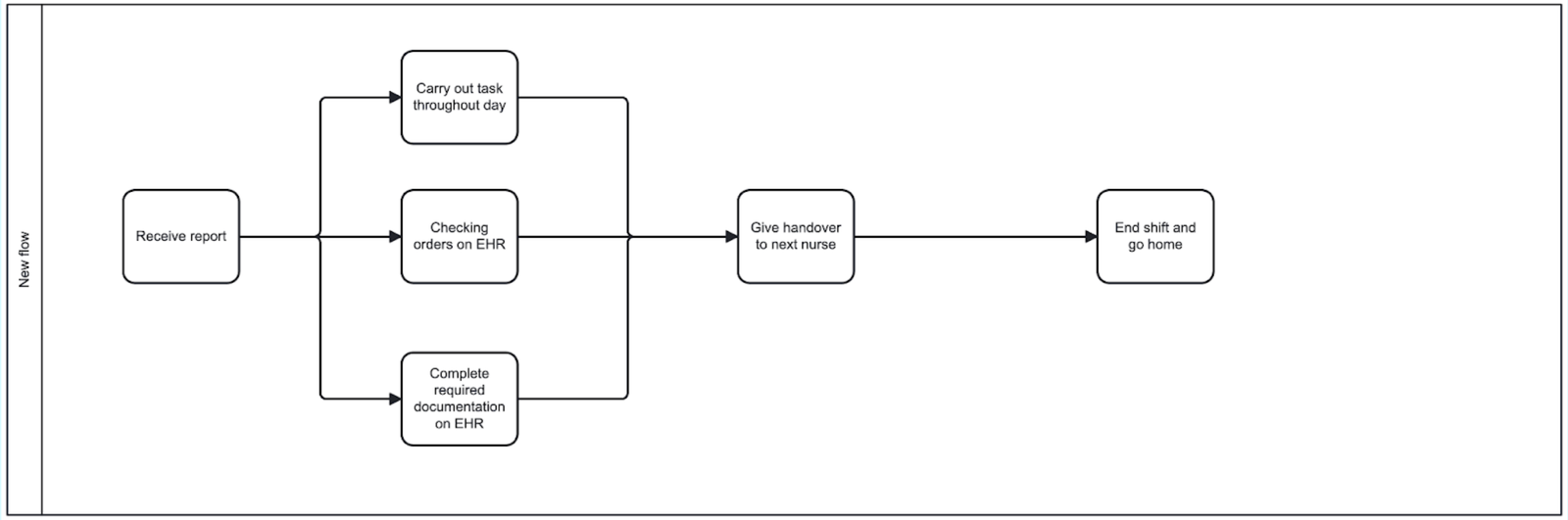
Lack of integration

PROTOTYPE


IPASS Nursing Handover Report
Standardized patient handover using IPASS methodology
Completion: 3/6 sections
Autosaved just now
Save Report
Save & Continue Later
Export
Clear
TRANSFER OF ACCOUNTABILITY
History of Presenting Illness
Chief Complaint & History
Describe the patient's chief complaint and relevant history...
Past Medical History
Diabetes mellitus
Hypertension
Coronary artery disease
Chronic kidney disease
COPD/Asthma
Heart failure
Atrial fibrillation
Stroke/CVA
Cancer history
Past Surgical History
Appendectomy
Cholecystectomy
Coronary artery bypass
Hip/Knee replacement
Hernia repair
Cardiac catheterization
Colonoscopy
Cataract surgery
Hysterectomy

Revised Workflow

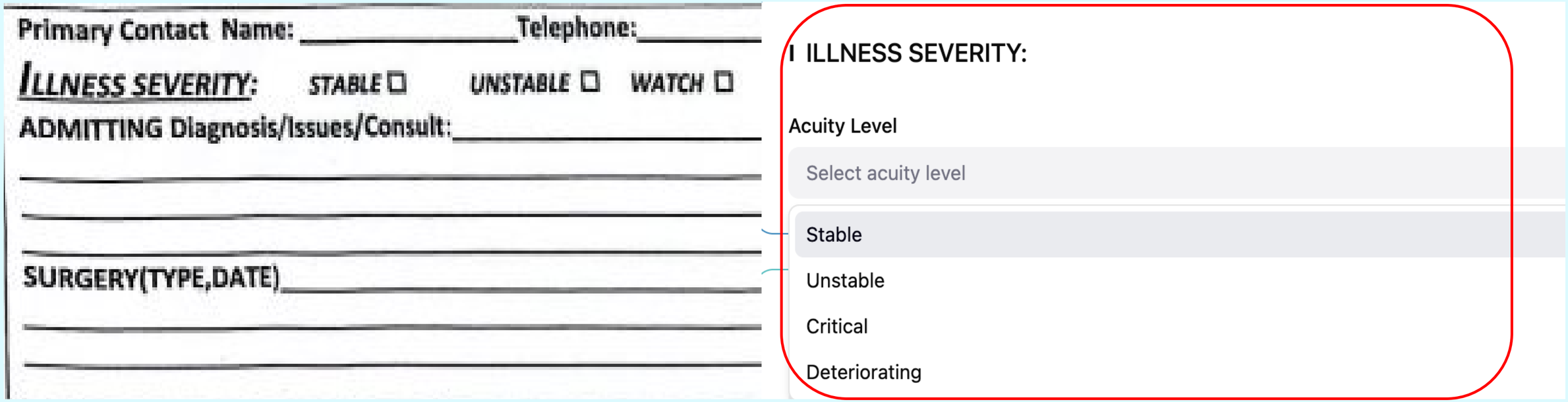




Human-Computer Interaction Principles

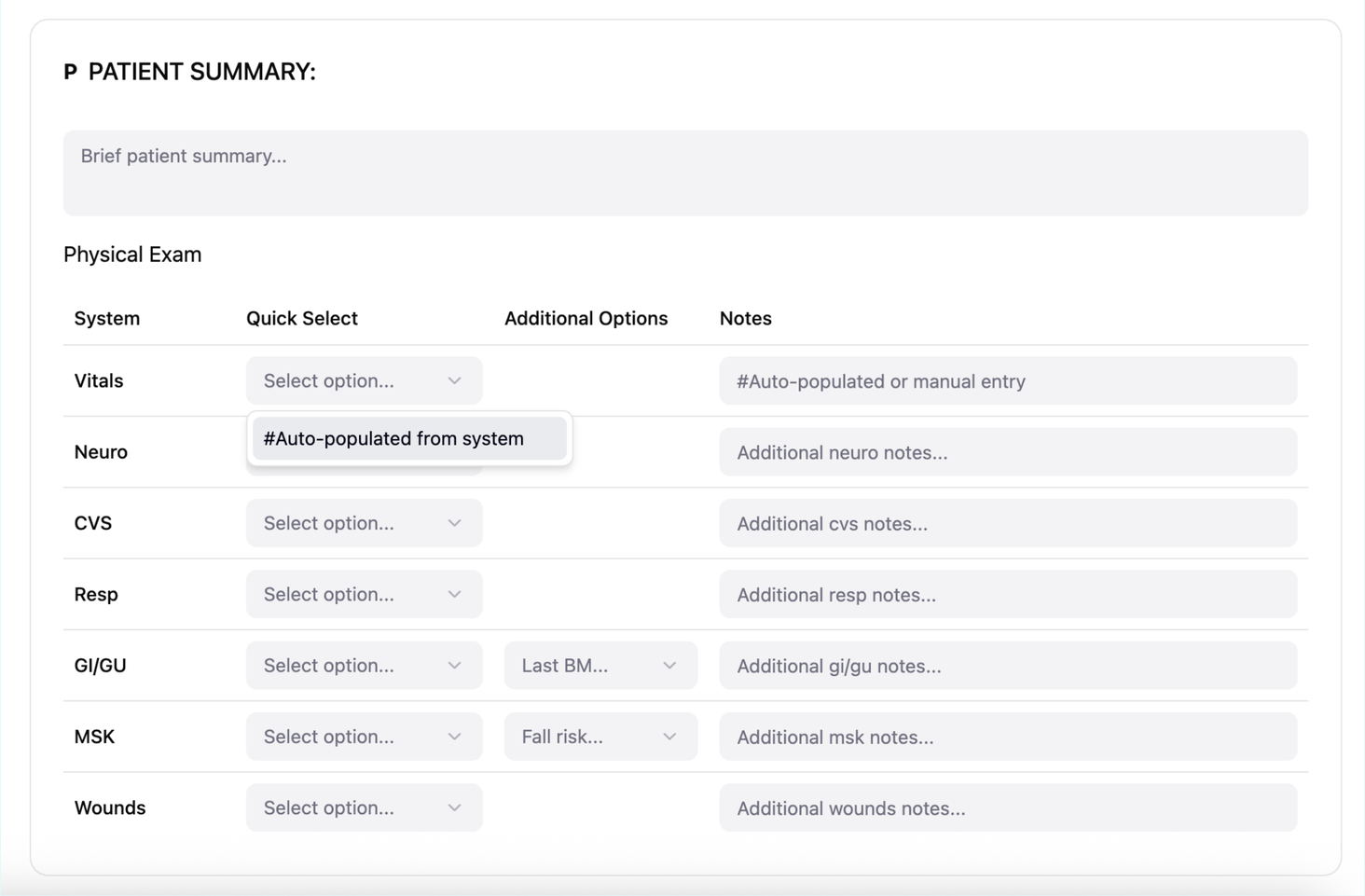
1. Strive for Consistency

- Same components when transitioning from the old form to the new digitalized form for standardization



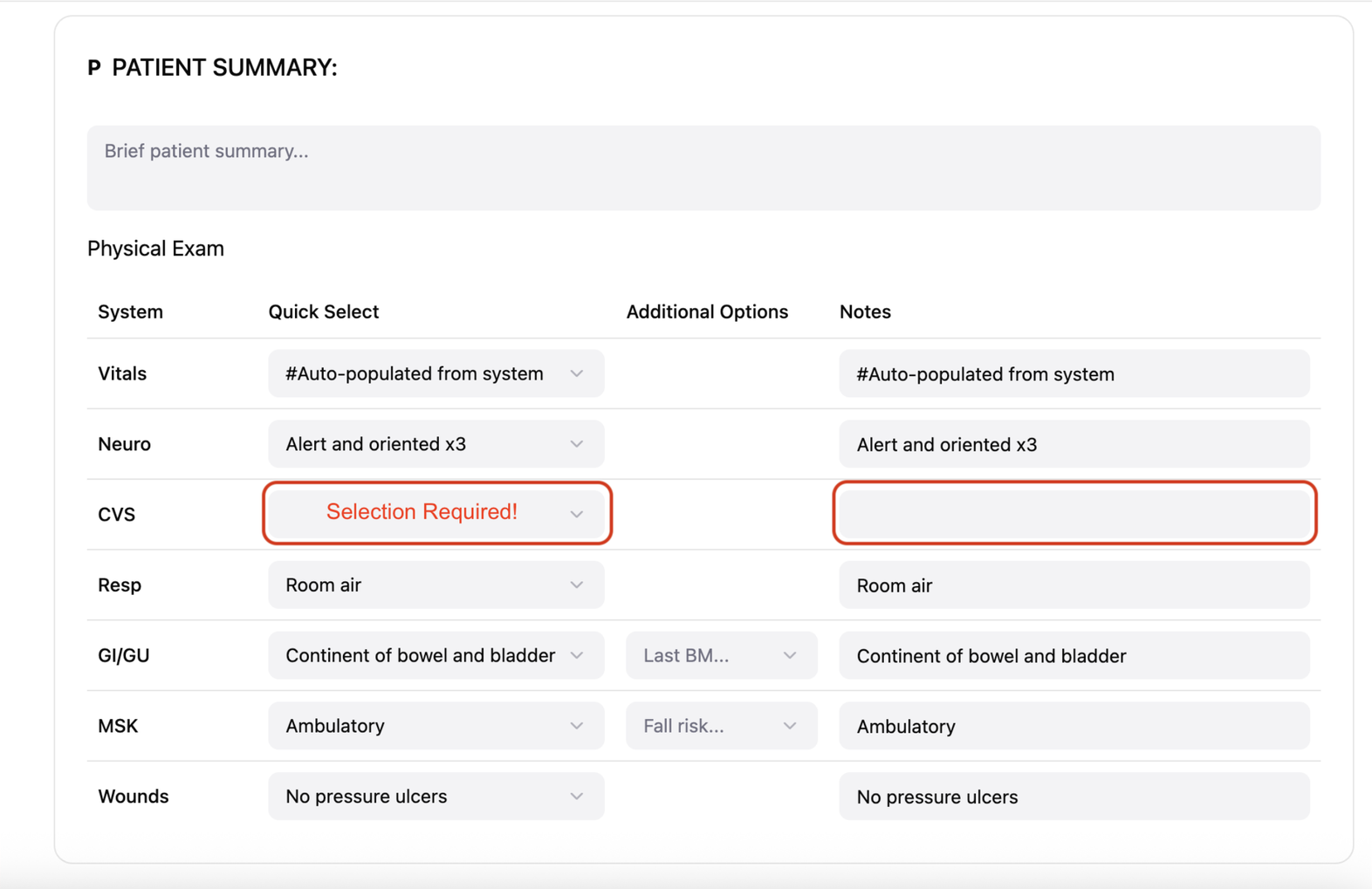
2. Enable Frequent Users to Use Shortcuts

- Auto-populate information (E.g. Past medication history), can be directly pulled from the EHR
- Shortcut tab in EHR to form



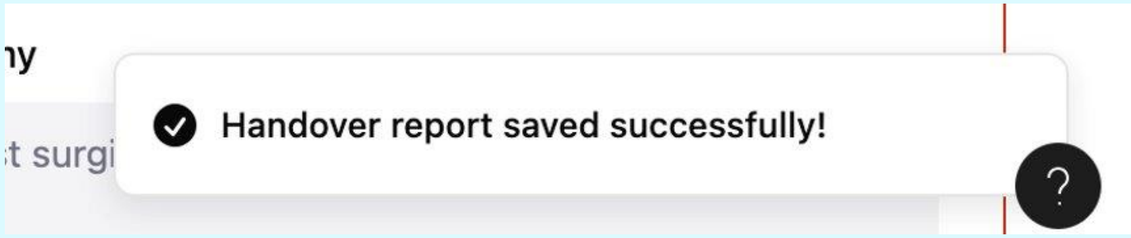
3. Offer Informative Feedback

- Prompts you to fill in required components/highlight the required component



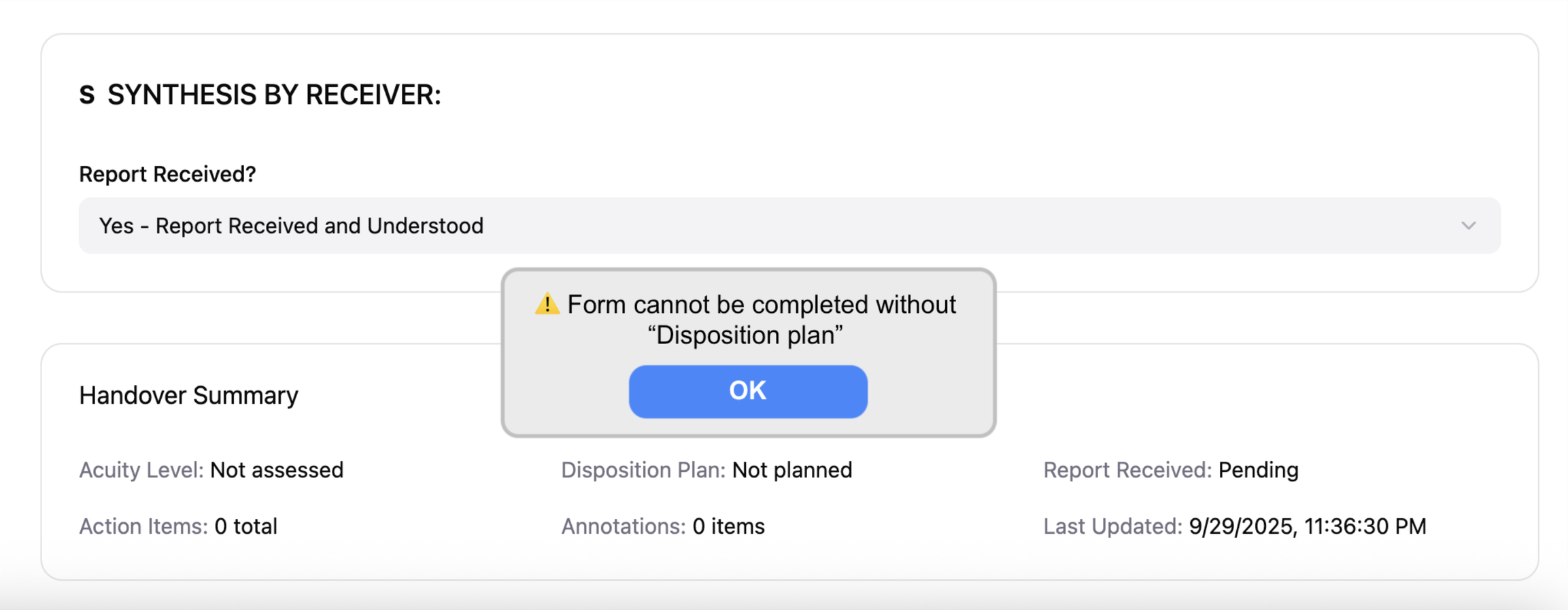
4. Design Dialog to Yield Closure

- Generates pop-up : “Response saved”
- It will automatically drop from your worklist when completed



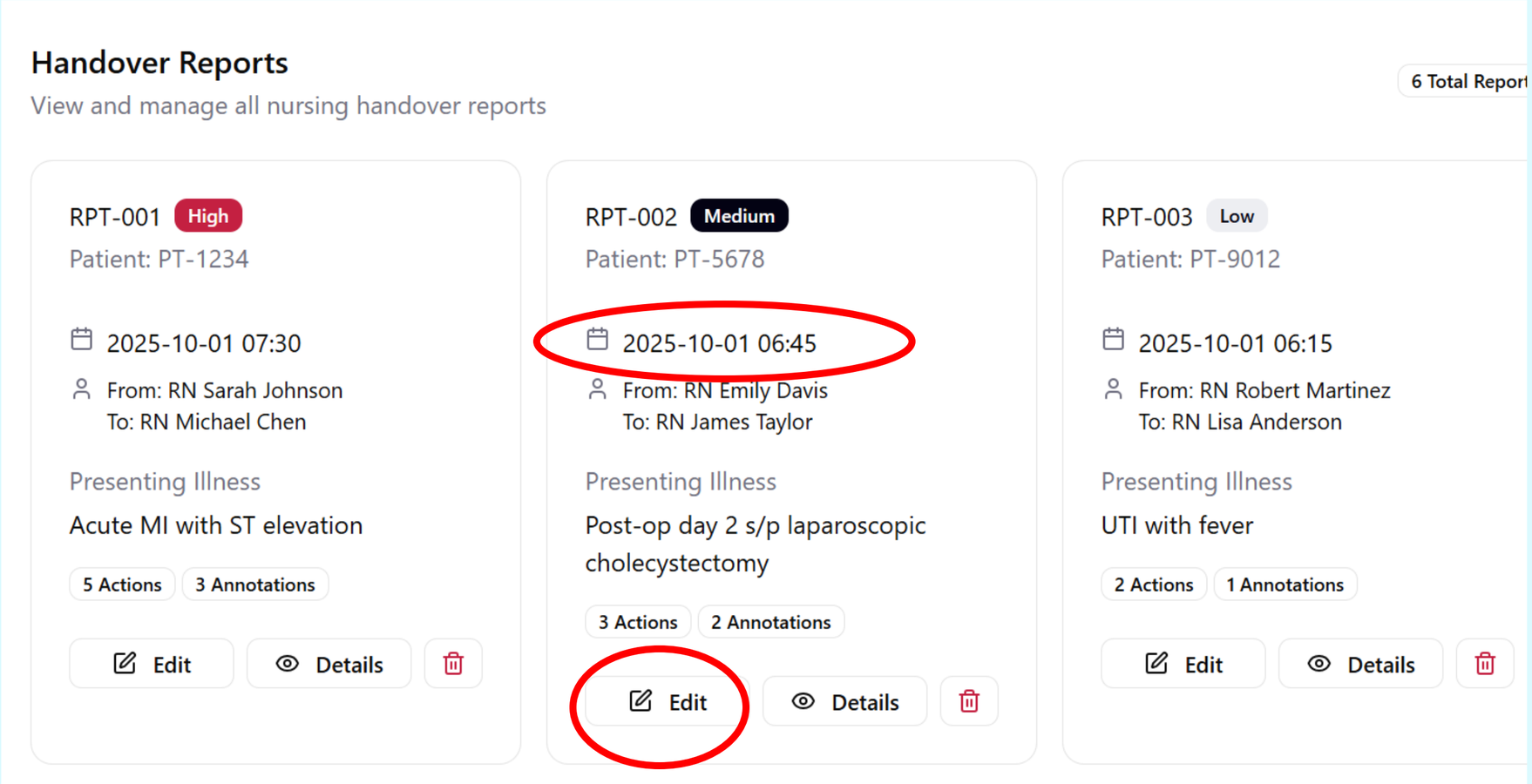
5. Offer Simple Error Handling

- Explain in detail what the error messages are



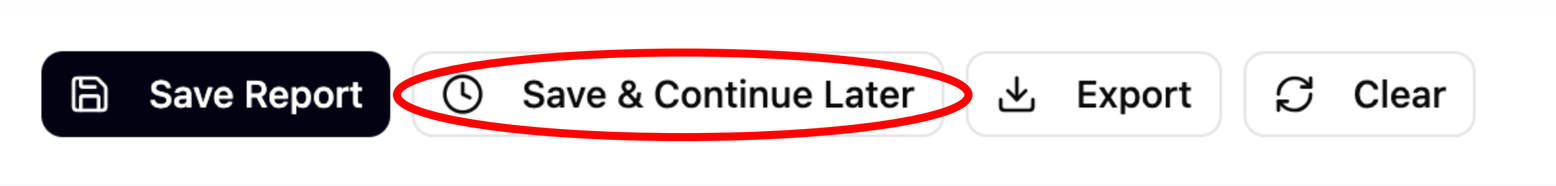
6. Permit Easy Reversal of Actions

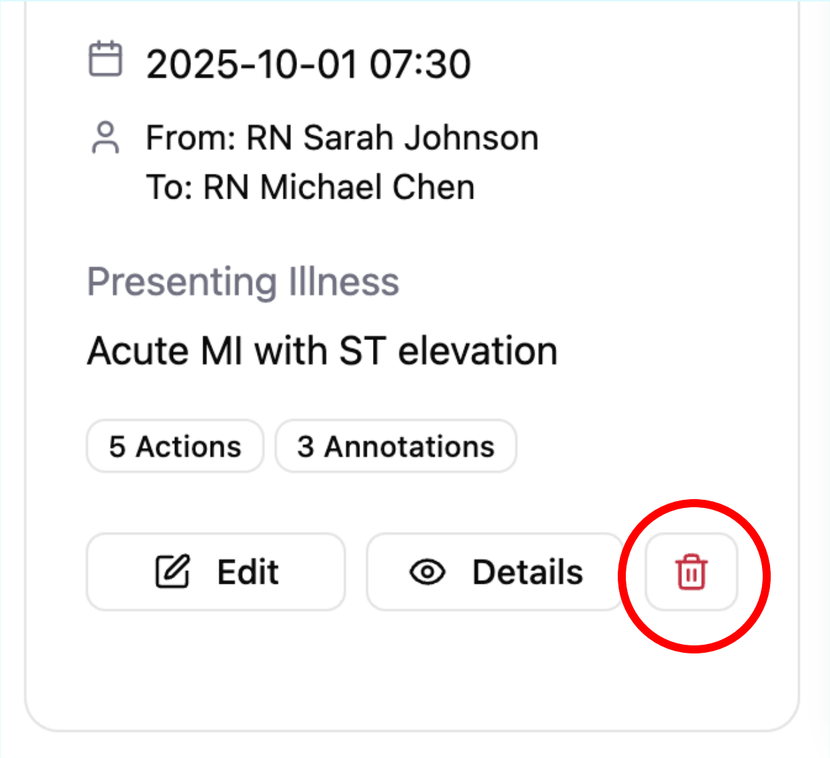
- Changes can be easily made but will be logged with timestamp & user ID
- Audit trail for accountability



7. Support Internal Locus of Control

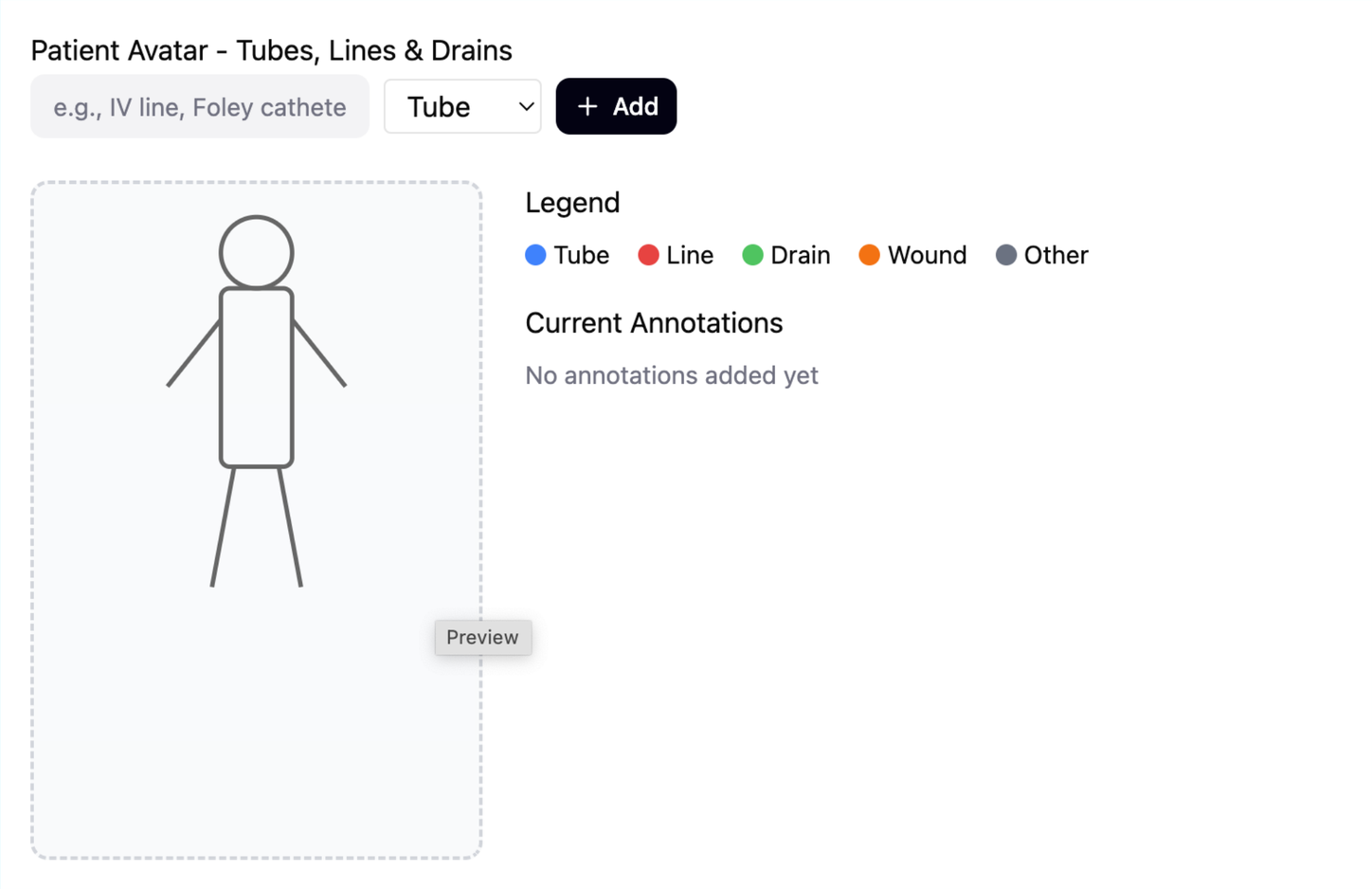
- Exit and return at any time - can save draft
  - System adapts to user needs



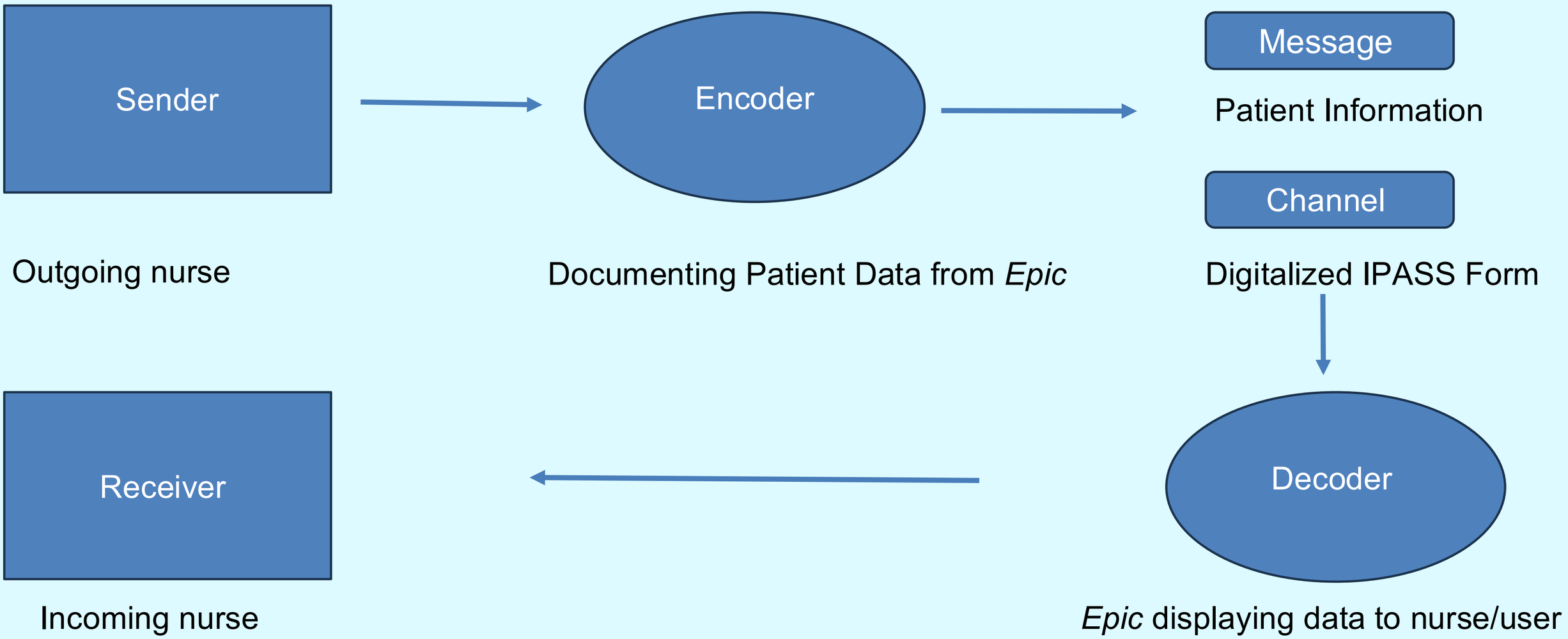


8. Reduce Short Term Memory Load

- Auto-populated vitals, medications, allergies
- Visual patient avatar (vs. text descriptions)
- No manual transcription required

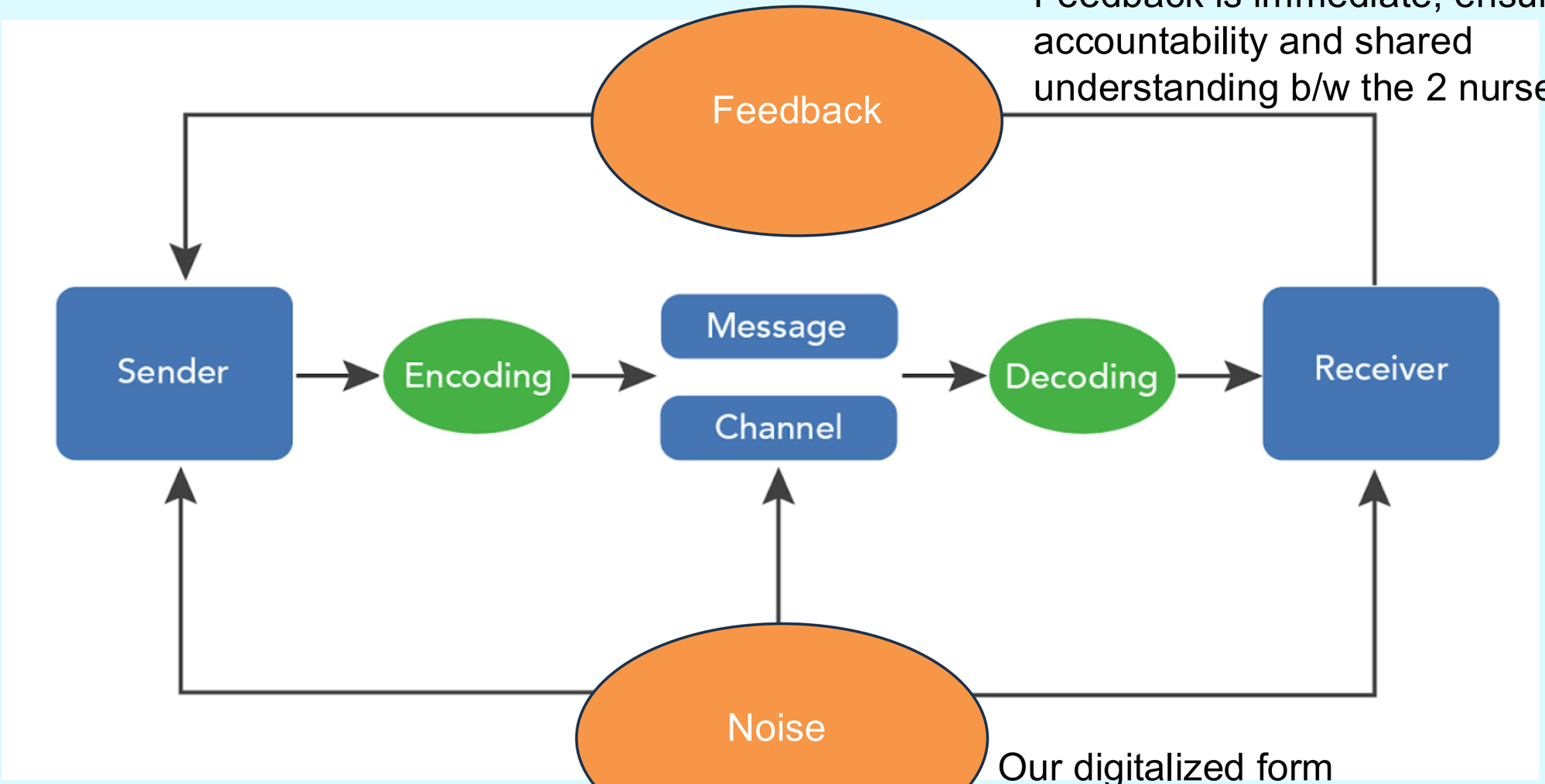


Shannon-Weaver Model



Shannon-Weaver Model

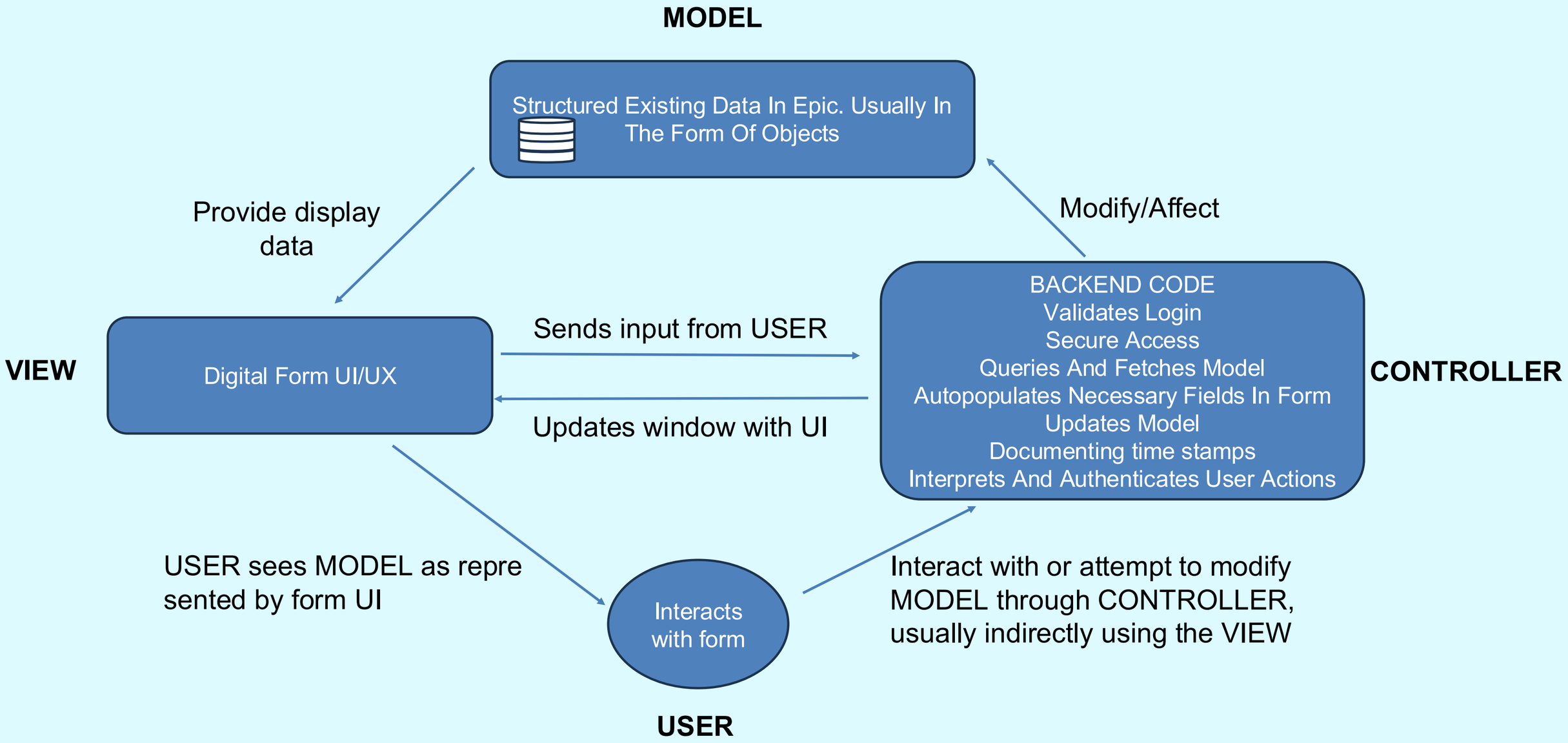
Feedback is immediate, ensures accountability and shared understanding b/w the 2 nurses



Our digitalized form REDUCES impacts from the noises

Software Architecture

MVC MODEL



Current Cost

Item	Calculation	Estimate
Nurse Time	6 forms x 30 minutes x $45/hr x 365 days	$49,275/year
Paper & Printing	6 forms/day x $1/page x 365 days	$2,190/year
Total Current Cost	-	**$51,465/year**

Future Cost

Item	Calculation	Estimate
Nurse Time	6 forms x 10 minutes x $45/hr x 365 days	$16,458/year
Paper & Printing	eliminated	$0
Total Future Cost	-	**$16,458/year**

Transition Cost

Item	Calculation	Estimate
Epic Customization Build	$50/hr* x 200 hours	$10,000
Nurse Training	42 nurses x 30 minutes x $45/hr	$945
Total Transition Cost	-	**$10,945**

*$100,000/2000 hours = $50/hr

	1 Year	3 Years
Cost of current process	$51,465	$154,395
Cost of future process	$16,458	$49,374
Cost of technology, training (one time cost)	$10,945	$10,945
Subtotal	$27,403	$60,319
Net savings	**$24,062**	**$94,076**

- Reduces errors
- Streamlines workflows
- Scales easily across other units

Any questions?

Thank you!

Link to Prototype: https://www.figma.com/make/VkbTP6b3QXEzBrNcuPgSi9/Healthcare-Handover-Form?node-id=0-1&t=A74g2kr7JsrpuKcA-1